

11-14 Sound and Waveforms

Prerequisites

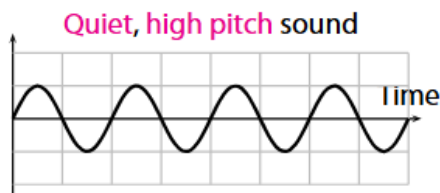
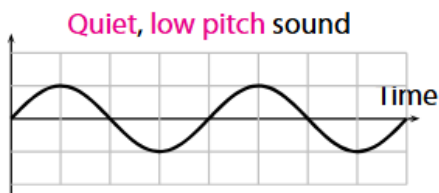
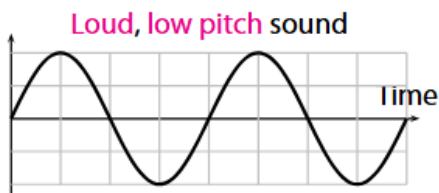
It is recommended that students have completed the **Waves** worksheet before this one, however this is not essential. However if students have not completed the Waves and Sound sheet, then they will need to be taught separately what **amplitude** means.

No prerequisites

The Main Points

- Sound waves and how to detect and view them
 - When sound travels, the particles oscillate in the material it is traveling through.
 - The time taken for a particle to complete one forward-and-backward motion is called the **time period**.
 - When sound reaches our ears or reaches a microphone, it makes a sensitive surface oscillate, and this is detected. When this is the ear, it is heard as sound.
 - When a microphone is connected to a data-logger or to an oscilloscope, a wave is seen on the screen. This shows the wave in one place as time passes.

- Sound and measurement



- Loud sounds have a large amplitude
 - Quiet sounds have a small amplitude.
 - Low pitched sounds have a long time period
 - High pitched sounds have a short time period.
- Frequency
 - Frequency is the number of complete oscillations each second.
 - High pitched sounds, with short time periods, have higher frequencies.
 - The frequency is calculated from $\frac{1}{\text{time period}}$ as this works out how many time periods there are in one second.
 - Frequency is measured in hertz (Hz).
 - High frequencies are measured in kilohertz (kHz) where 1 kHz = 1000 Hz.
 - Humans can usually hear sounds between 20Hz and 20kHz = 20 000 Hz.
 - Sounds above 20 kHz are called **ultrasound**

Teacher Quarter Briefing

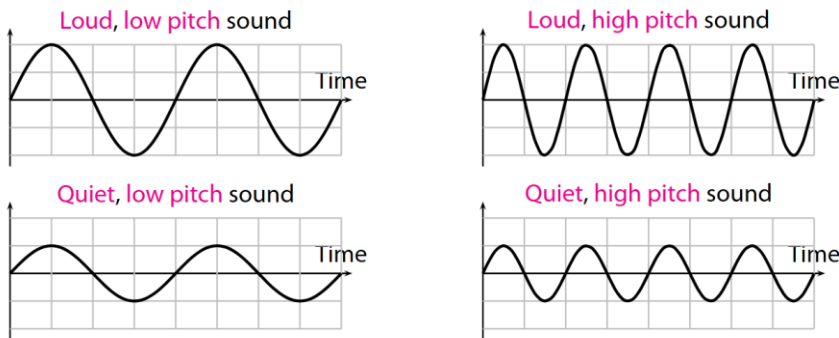
- Introduction: <https://youtu.be/SDjFQC0vntA>
- Practice: https://isaacscience.org/question_decks#itsp_waveform
- Review: https://youtu.be/jcF_2t0CbEQ

Class Question Notes

The worksheet can be printed either in full, or in cloze text form (where the red text is missing, and students can complete these blank spaces after class discussion). The online version of the notes requires the appropriate text to be dragged to the right place in the sentences.

[Shallow learning gradient online assignment](#) - q1,3,4,5,6,7,8,9,10

[Steeper learning gradient online assignment](#) - q1,2,4,5,6,7,8,9,11,12



1. In this question we ask students to compare the graphs and note
 - a. loud sounds look 'taller' (they go up and down further) on the graph than quiet sounds
 - b. high sounds have more wiggles on the graph than low pitched sounds
 - c. A short time period means that each wave (or wiggle) takes up little room on the graph, so there is room for lots of wiggles, and this is a high pitched sound.
 - d. A large amplitude means that the shape on the graph goes up and down more than a small amplitude. So a large amplitude sound is going to be loud.
2. Here students should draw their own graphs which show
 - a. curves which stay near the horizontal axis (v small amplitude) but with more than four complete waves so that it is higher than the sounds drawn in the graphs from q1.
 - b. curves which go higher and lower than the existing curves (very large amplitude) but with fewer than two whole waves on the graph so that the sound is lower than the previous sounds.
3. Here students rank graphs
 - a. from quiet to loud: start with the one with the smallest amplitude (B) and work up to the one with the largest amplitude (C) which has the tallest peaks
 - b. from low to high: start with the one that is least wiggly (fewest waves: D, which has barely more than half a wave) and work up to the one with the most waves (B which has 4 complete waves).
4. Here students note that if D is the lowest sound in q3, it has the lowest frequency.
5. Students complete the blanks with words from a list describing the frequencies we can hear.
6. Students take times in milliseconds and express them in seconds. You may wish to show them your own preferred method for this (remind them it is just like taking distances in metres and putting them in millimetres). Some students may just want to think in terms of multiplying by 1000, others might prefer to move the decimal point to the right 3 times.
7. Students take times in seconds and express them in milliseconds. You may wish to show them your own preferred method for this (remind them it is just like taking distances in millimetres and putting them in metres). Some students may just want to think in terms of dividing by 1000, others might prefer to move the decimal point to the left 3 times after adding enough leading zeroes to make this possible (e.g. with (b) writing 3 ms as 0003.0 ms before moving the decimal point 3 times to the left to give 0.0030s).
8. Students work out the time period by measuring the time between, for example, one peak (at 50ms) and the next (at 100ms) and taking the difference: 50ms.

9. In this question, students are encouraged to explore the number pattern between a time period and the frequency which goes with it. The first example is a time period of 0.05s. There are $1/0.05 = 20$ of these in one second, so the frequency is 20Hz.
10. This question has more parts. Students write the time period as $5\text{ms} = 0.005\text{s}$, and then do $1/0.005\text{s}$ to get the frequency in hertz.
11. In this question, students use the method of q9 to work out the frequency of the 20Hz wave in q8.
12. In this question, students practise working out frequencies from time periods, but have to give their answers in kilohertz (kHz). Most students will want to work steadily, for example in (d) writing time period = $0.1\text{ms} = 0.0001\text{s}$ so frequency = $1/0.0001\text{s} = 10\,000\text{Hz} = 10\text{ kHz}$. Others might notice that if you divide 1 by the time period in milliseconds, you get the frequency in kilohertz.

Homework Question Notes

These questions have a very similar form to the questions in the class task, so students can refer back to their earlier answers to help.

[Shallow learning gradient online assignment](#) - q1,2,3,4,5,6,7,8,9,11

[Steeper learning gradient online assignment](#) - q1,2,3,5,6,7,8,10,11,12

1. In this question, students rank waves
 - a. from loudest (A, as it's peaks are tallest) to quietest (B),
 - b. from highest (D, it is the most wiggly and has the most complete waves) to lowest (A).
 - c. The wave with the shortest time period is the one where the most waves will fit on the graph, so it is the highest pitch wave.
2. Here students need to know that high frequency means short time period.
3. This question asks students to assign frequencies from a list to fill in the blanks in text. Smoke alarms need to be high and annoying, so are usually about $3000\text{Hz} = 3\text{kHz}$. The E on a bass guitar is low, but just about audible (41 Hz).
4. Here, just as in class question 6, students write times in milliseconds, having been given them in seconds.
5. Here, just as in class question 7, students write times in seconds, having been given them in milliseconds.
6. Students look at a graph and read off the time period - for example the time elapsed from one peak (0ms) to the next (100ms). Note that the question wants the answer to be given in seconds, so they need to express 100ms as 0.1s.
7. This question looks at the link between time period and frequency just like class q9.
8. Students calculate a frequency in Hz from a time period in ms.
9. Students calculate the frequency of the wave in q6.
10. Students practise calculating frequencies in kHz from time periods in ms.
11. Students match graphs to descriptions of sounds.
12. This question shows two graphs - one of a sound suddenly going up in pitch (the one which gets more wiggly as the line goes to the right - A) and one going suddenly down in pitch (the one which gets less wiggly - B). They have to match one of the waves to a situation given in the question.

Extension questions from 'Step Up to GCSE Physics':

https://isaacscience.org/question_decks#phys_step_up_35

https://isaacscience.org/question_decks#phys_step_up_37